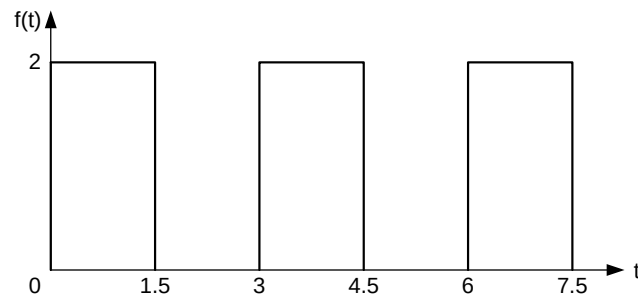


Problem Sheet 8

1.

(a) The figure shows a sequence of pulses. Write $f_1(t)$, the form of the first pulse, in terms of a super-position of two unit steps. Write the Laplace transform $F_1(s) = \mathcal{L}\{f_1(t)\}$.



(b) Give $f_2(t)$, the form of the second pulse, in terms of $f_1(t)$ with a time-shift. Write the Laplace transform of the second pulse.

(c) Show that the Laplace transform of the first three pulses combined can be written as

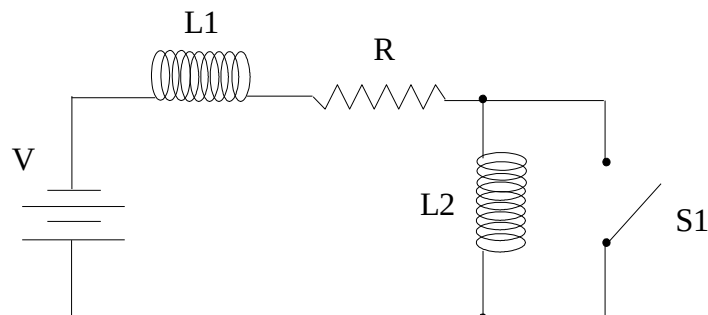
$$F(s) = F_1(s)(1 + e^{-3s} + e^{-6s}).$$

(d) In the case that the pulses repeat recurrently, show that

$$F(s) = \frac{2}{s} \left(\frac{1}{1 + e^{-1.5s}} \right).$$

2. SEMINAR QUESTION I

(a) A circuit containing a resistor and two inductors is constructed as shown below.



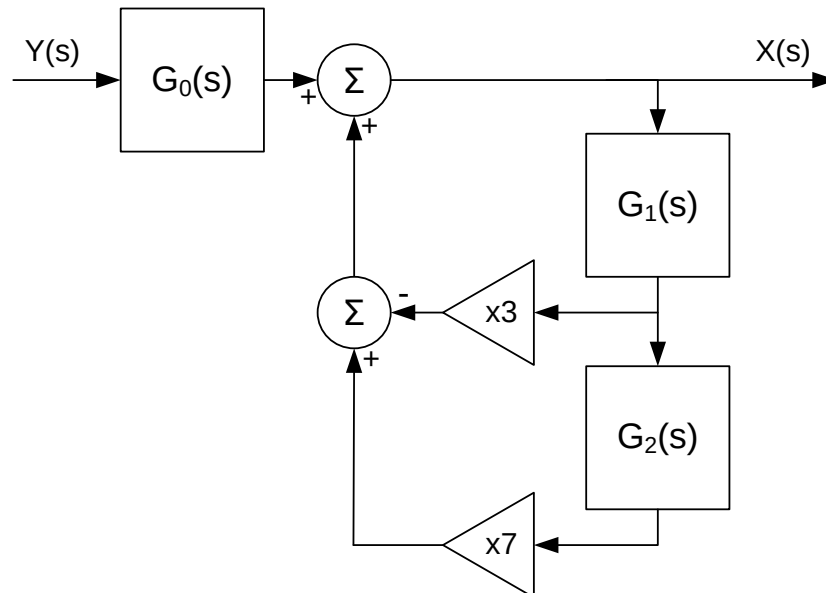
Initially, the switch S_1 is closed (shorting out inductor L_2). The switch is opened at time $t = 0$, increasing the effective inductance of the circuit from L_1 to $(L_1 + L_2)$. Write down a simple expression for the steady state current that flows through the circuit at a time $t < 0$ before the opening of the switch.

(b) Write down a simple expression for the steady state current in the limit $t \rightarrow \infty$ after opening the switch.

(c) Give the differential equation relating the current and voltage in the circuit for $t > 0$. Use the Laplace transform to solve this equation and derive an expression for the time domain current for all time $t > 0$.

3. SEMINAR QUESTION II

(a) The system diagram below contains three functional blocks and two ideal amplifiers of gain $\times 3$ and $\times 7$. It contains both positive and negative feedback. Derive an expression for the transfer function of the system in terms of $G_0(s)$, $G_1(s)$ and $G_2(s)$.



(b) If G_0 has a time-domain impulse response of $24te^{-2t}$, G_1 is a unity-gain differentiator, and G_2 is a unity-gain integrator, express the system transfer function in terms of s .

(c) Determine whether the system as a whole will be stable. Will it oscillate?

(d) Find the time-domain response to a unit-step $u(t)$ applied at the input.

Numerical Answers and Hints

1 (a) The pulse decomposes into a unit-step plus a time-shifted unit-step. For the latter, make use of the 'frequency-shift' property of the Laplace transform $\mathcal{L}\{f(t - \lambda)u(t - \lambda)\} = e^{-s\lambda}F(s)$. For part (d) make use of the Maclaurin expansion of $(1 - x)^{-1}$ and the geometric series S_n (formula sheet).

2 Note that in the steady-state condition the inductor plays no part in the circuit behaviour (it behaves like a simple wire with zero resistance for a DC current). Hence the answer to (a) and (b) is simple (and both answers are the same) (c) Having found the current immediately *before* the closing of the switch in (a), find the current immediately *after* the closing of the switch using conservation of magnetic flux.